Original Article

Surgical Management of Posterolateral Tibial Plateau Fractures: A Comparative Study of Direct Lateral and Posterolateral Approaches

Reza Noktehsanj[1] | Ali Nami[1,*] | Aziz Kamran[2] | Amin Najjar Khodabakhsh[3]

[1]*Department of Surgery, School of Medicine, Fatemi Hospital, Ardabil University of Medical Sciences, Ardabil, Iran*
[2]*Department of Community Medicine, School of Medicine, Ardabil University of Medical Sciences, Ardabil, Iran*
[3]*School of Medicine, Ardabil University of Medical Sciences, Ardabil, Iran*

Use your device to scan and read the article online

Citation R. Noktehsanj, A. Nami, A. Kamran, A. Najjar Khodabakhsh, **Surgical Management of Posterolateral Tibial Plateau Fractures: A Comparative Study of Direct Lateral and Posterolateral Approaches.** *Eurasian J. Sci. Technol.*, **2026**, *6*(1), 74-83.

https://doi.org/10.48309/ejst.2026.545126.1264

CC BY

Article info:
Received: 2025-09-04
Accepted: 2025-10-15

ID: EJST-2509-1264

Keywords:
Tibia, Bone Fractures, Fracture Fixation, injury, Knee , Surgery.

ABSTRACT

Posterolateral tibial plateau fractures are challenging injuries of the knee joint, which is essential for weight-bearing and stability. These fractures typically occur due to high-energy trauma and involve complex surgical management due to the proximity of neurovascular structures and surrounding ligaments. This study aims to compare the clinical outcomes of two surgical approaches—direct lateral and posterolateral—for the fixation of posterolateral tibial plateau fractures. A descriptive was conducted on 54 patients who were evenly divided into two groups. One group underwent the direct lateral approach, while the other received the posterolateral approach. The primary outcomes measured included the International Knee Documentation Committee (IKDC) score, Lysholm score, range of motion, neurovascular injury, fixation success, and quality of joint reduction. Patients treated with the direct lateral approach showed significantly higher IKDC scores (92 ± 2.4) compared to the posterolateral group (90 ± 1.3), with $p < 0.001$. Similarly, the Lysholm scores were better in the lateral group (96 ± 2.4) versus the posterolateral group (88 ± 1.5), $p < 0.001$. The range of knee motion was also superior in the direct lateral group (130° vs. 125°, $p < 0.001$). Neurovascular injuries were fewer in the direct lateral approach. The modified direct lateral approach offers better clinical outcomes compared to the posterolateral approach, minimizing neurovascular risks and improving joint reduction and fixation. This method provides a safer and more effective surgical option for managing posterolateral tibial plateau fractures.

Introduction

The knee joint is one of the three primary weight-bearing joints in the lower limbs, and injuries to this joint can result in significant instability, pain, and impaired function. Proximal tibial plateau fractures are among the most critical traumatic knee injuries [1,2]. These fractures account for roughly 1% of all fractures and were historically known as "bumper" or "fender" fractures, typically caused by low-energy trauma such as a pedestrian impacted by a car bumper. However, modern cases more commonly arise from high-energy trauma like motor vehicle collisions or falls from heights and may be accompanied by other injuries in the same or different anatomical regions [3,4].

*Corresponding Author: Ali Nami: ali.nami.damirchi@gmail.com

Posterolateral tibial plateau fractures, involving the posterolateral quadrant, make up 7–14% of proximal tibial fractures. Recent data suggest their incidence could be as high as 44%, with 36% classified as AO Type B and 54% as Type C fractures [5-7]. These fractures typically result from direct axial compression combined with valgus or other forces, causing impact from femoral condyles onto the tibial plateau. The characteristics of the injury—such as force magnitude, direction, and knee position—drive the fracture pattern and displacement extent. Patient age and bone quality also influence fracture types; for example, older patients more often sustain Schatzker type III compression fractures, whereas younger individuals more frequently experience fractures involving ligament injury [1,8,9].

No universally accepted standard treatment exists for tibial plateau fractures, with various methods documented [10,11]. Treatment goals include anatomical joint surface reduction, stabilization of bone and ligaments, preservation of knee mechanical alignment, soft tissue protection, and restoration of joint mobility. Inadequate management may lead to pain, instability, deformity, restricted motion, and eventual osteoarthritis [12,13]. Non-surgical management can be appropriate for non-displaced or minimally displaced fractures, with acceptable joint depression ranging between 2 and 8 mm for conservative treatment. Surgical fixation of posterolateral tibial plateau fractures is challenging due to the complex anatomy, including the fibular head, ligament attachments, and critical neurovascular structures surrounding the posterolateral quadrant. Thus, controversy remains over the optimal surgical approach.

Several studies have examined surgical techniques for this fracture type.

This study aims to compare the clinical outcomes of two surgical approaches—the direct lateral versus posterolateral—for fixation of posterolateral tibial plateau fractures. It is hypothesized that the direct lateral approach will offer superior functional outcomes, a lower rate of neurovascular complications, and more effective fixation. This research addresses a literature gap, as no comprehensive comparative study currently exists for these approaches in managing posterolateral tibial plateau fractures.

Materials and Methods

Study Design

This case series was conducted at the Department of Orthopedics, Ardabil University of Medical Sciences, between 2020 and 2022. The study focuses on comparing two surgical approaches—posterolateral and direct lateral—for the treatment of posterolateral tibial plateau fractures.

Study Population

The study included patients with posterolateral tibial plateau fractures who were admitted to Fatemi Hospital in Ardabil and Taleghani Hospital in Tehran during 2020 and 2021. The inclusion criteria were:

- Patients aged between 18 and 70 years,
- Fractures involving the posterolateral tibial plateau,
- Consent to participate in the study.

Patients were excluded if they had a prior peroneal nerve injury or previous open knee surgery.

A total of 54 patients were divided into two groups:

- **Group A**: Underwent surgery using the posterolateral approach,
- **Group B**: Treated with the direct lateral approach.

Sample Size Calculation

The sample size based on 95% confidence interval and 80% power, was calculated using G-Power software based on the formula and necessary information. This resulted in 27 patients per group (54 patients total).

Data Collection

After obtaining informed consent, demographic data were recorded, including age, gender, and the affected side. Hospital-related data such as length of hospital stay, surgery duration, incision

length, and intraoperative blood loss were also collected.

Postoperative data were gathered based on functional scores at the end of the follow-up period. These functional scores included:

1. Lysholm Knee Score: This score is based on eight domains—squatting, locking, pain, stair climbing, support, instability, swelling, and overall functionality—ranging from 0 to 100. Scores were categorized as follows:

- 91–100: Excellent
- 84–90: Good
- 65–83: Fair
- ≤64: Poor.

2. International Knee Documentation Committee (IKDC) Score: This tool assesses knee symptoms (seven items), function (two items), and sports activities (two items). Scores range from 0 to 100, with higher scores indicating better performance.

3. Tegner Activity Scale: This scale ranks physical activity on a scale of 0 (indicating disability) to 10 (elite sports performance).

Fracture healing, infection at the surgical site, and hardware-related symptoms were evaluated during follow-up clinical examinations and radiographic assessments.

Surgical Techniques

Direct Lateral Approach

The patient was placed in a supine position with the leg hanging freely from the table. A 10 cm skin incision was made starting 3 cm proximal to the joint line and extending distally along the fibula. The common peroneal nerve was identified and carefully retracted. The iliotibial band was elevated, and the lateral collateral ligament (LCL) was exposed. Joint depression was reduced using a bone tamp under fluoroscopic guidance. A bone graft was used to fill the void, and the fracture was stabilized with rafting screws (Figures 1-3).

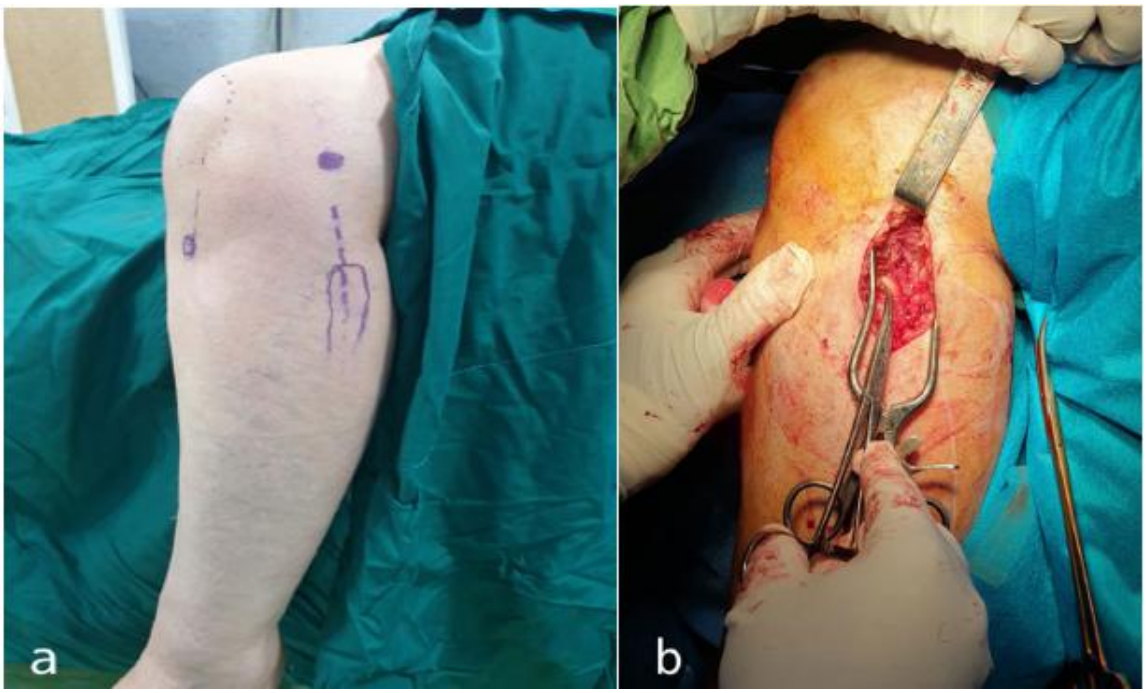


Figure 1 Intraoperative positioning and skin incision for the direct lateral approach. (a) The patient is positioned supine with the leg hanging freely from the edge of the operating table. (b) The incision follows the lateral side of the leg, avoiding the risk

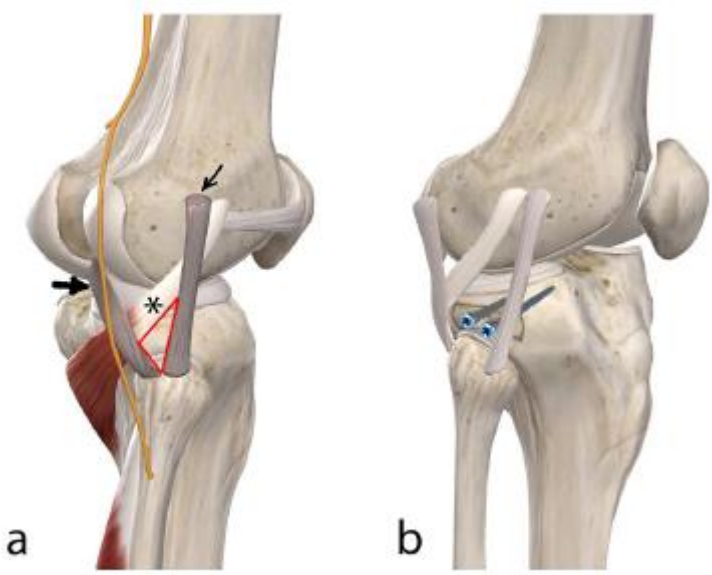


Figure 2 a. The reduction of a posterior tibial plateau fracture achieved through a safe surgical window, with anatomical landmarks such as the LCL (thin arrow), popliteal tendon (star), and popliteofibular ligament (thick arrow). b. Fixation was performed using rafting screws placed towards the intact posterior section of the tibial plateau, securing the graft in position



Figure 3 A 24-year-old male with posterior tibial plateau fracture and fibular head fracture treated using the technique described in this study, along with an ACL tuberosity fracture. (a) Early postoperative X-ray and (b) X-ray after 3 months

Posterolateral Approach

In posterolateral approach [14-17], patients were placed in a prone position. A 10 cm incision was made along the medial border of the biceps femoris tendon. Subcutaneous tissue and the popliteal fascia were dissected to expose the interval between the biceps femoris and lateral gastrocnemius muscles. The peroneal nerve was identified and retracted laterally. The gastrocnemius origin was released, and the soleus was elevated to avoid injury to the anterior tibial artery. Joint reduction was achieved through a posterior approach, and a T-shaped precontoured plate was used for fixation (Figures 4,5).

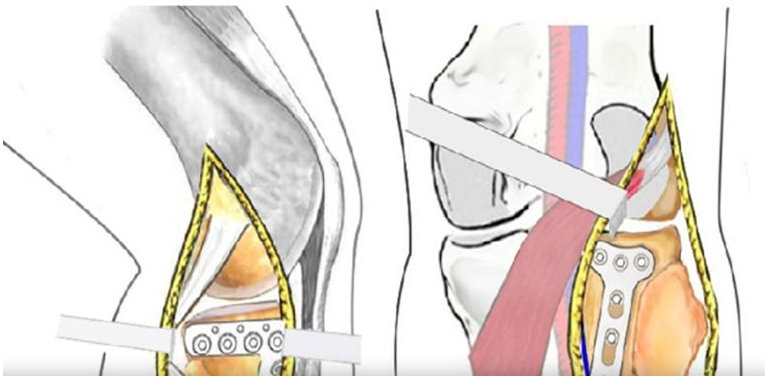


Figure 4 Reduction and fixation using the posterolateral approach. (a) Reduction of the posterior tibial plateau through the safe zone for neurovascular anatomy. (b) Insertion of pre-contoured T-shaped plate and fixation with rafting screws.

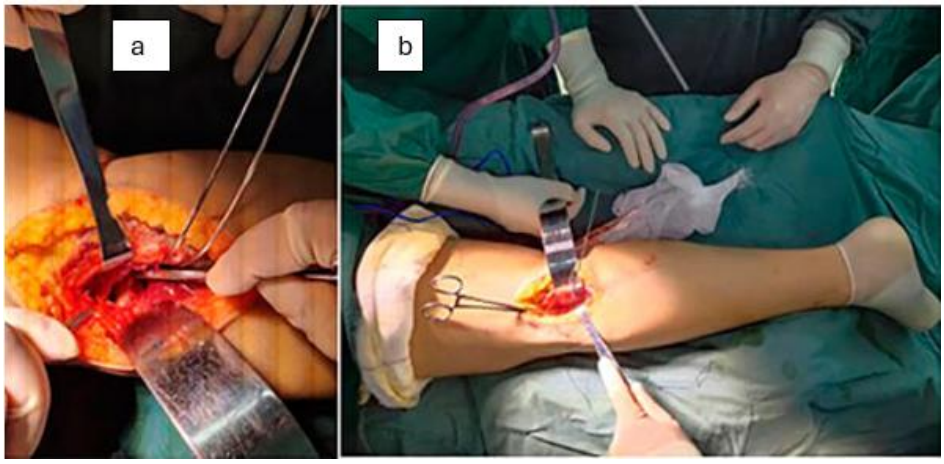


Figure 5 a) Skin incision in the posterolateral approach and b) exposure of the posterolateral tibial plateau

Postoperative Care

The same rehabilitation protocol was followed for both groups:

- Continuous passive range of motion (ROM) exercises were initiated 1 week postoperatively.
- Patients were non-weight-bearing for 6 weeks, followed by partial weight-bearing.
- Full weight-bearing was allowed after 10 weeks.

Radiographic and Functional Evaluations

Radiographic evaluations were performed using X-rays at preoperative, 2-week, 1-month, 3-month, and 12-month intervals postoperatively. CT scans were also obtained preoperatively and postoperatively to assess joint alignment and fracture healing.

Functional outcomes were evaluated preoperatively and at the final follow-up using the Lysholm, IKDC, and Tegner scales (Figure 6).

Statistical Analysis

All statistical analyses were performed using SPSS version 26.0 (IBM Corp., Chicago, IL, USA). Descriptive statistics were used to summarize demographic data, and comparisons between the two surgical approaches were analyzed using independent and paired **t-tests**. Continuous variables were expressed as mean ± standard deviation (SD), and a p-value of less than 0.05 was considered statistically significant.

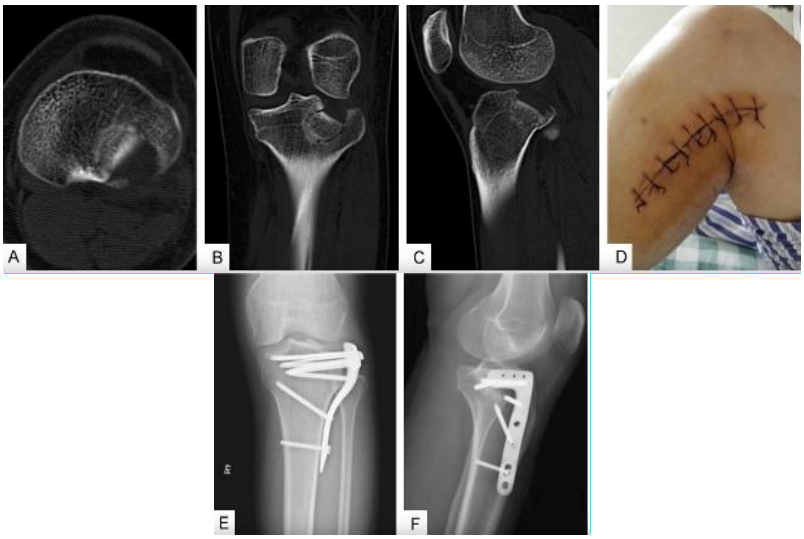


Figure 6 Preoperative and postoperative CT scans of a 29-year-old female patient with a posterolateral tibial plateau fracture. (a) Preoperative scan showing the extent of the fracture. (b) Postoperative radiograph showing satisfactory reduction and hardware placement

Ethical Considerations

The study adhered to ethical guidelines set by the National Medical Research Ethics Committee. Approval was granted with ethics code IR.ARUMS.MEDICINE.REC.1402.076. Written informed consent was obtained from all participants. Patient data were handled with confidentiality, and participants were free to withdraw from the study at any time. No additional diagnostic or therapeutic procedures beyond standard care were performed.

Results

In the study period of 2019-2021, 54 patients (25 females and 29 males) with an average age of 32.1 years (ranging from 18 to 70 years) were included. All patients had closed fractures. Each case was evaluated both preoperatively and postoperatively using X-ray radiography, CT scans, and 3D reconstructions. The causes of injury included road traffic accidents in 38 patients and falls from height in 16 patients. Out of the total patients, 50 had isolated posterolateral fractures, while 4 had both anterolateral and posteromedial fractures. The final follow-up was conducted 12 months postoperatively.

Table 1 compares the baseline demographic characteristics and preoperative data of the two groups (posterolateral and lateral approach). The mean age of the patients was 32.1 years, with 46.29% being female. The table shows that there was no significant difference between the posterolateral and lateral approach groups in terms of gender distribution (male/female: 13/14 vs. 16/11, p=0.53), age (30.36 ± 6.50 vs. 33.93 ± 4.33, p=0.90), and the side of the fracture (right/left: 15/12 vs. 10/17, p=0.45).

Table 1 Comparison of Preoperative characteristics between posterolateral and lateral approach groups.

Variable	Posterolateral group	Lateral group	*p*-value
Age (years)	30.36 ± 6.50	33.93 ± 4.33	0.90
Gender (%) Male	13 (48.1)	16 (59.25)	0.53
Gender (%) Female	14 (46.7)	11 (40.74)	0.53
Fracture Side (%) Right	15 (55.55)	10 (37.03)	0.45
Fracture Side (%) Left	12 (44.44)	17 (62.96)	0.45

The hospital data, including the length of hospital stay, duration of surgery, incision length, and intraoperative blood loss, are summarized in Table 2. The mean hospital stay for the posterolateral group was 5.42 ± 1.72 days, while for the lateral group, it was 4.35 ± 1.17 days, which was statistically significant (p=0.013). The mean operative time in the posterolateral group was 63.25 ± 4.29 minutes, significantly longer than the lateral group's 43.53 ± 3.89 minutes (p=0.001). The average incision length in the posterolateral and lateral groups was 155.80 ± 30.59 mm and 103.77 ± 40.28 mm, respectively, showing a significant difference (p=0.001). Blood loss during surgery was significantly higher in the posterolateral group (68.18 ± 6.37 mL) compared to the lateral group (43.16 ± 11.47 mL) (p=0.001).

Table 2 Comparison of hospital data between posterolateral and lateral approach groups

Variable	Posterolateral group	Lateral group	*p*-value
Length of hospital stay (days)	5.42 ± 1.72	4.35 ± 1.17	0.013
Operative time (minutes)	63.25 ± 4.29	43.53 ± 3.89	0.001
Incision length (mm)	155.80 ± 30.59	103.77 ± 40.28	0.001
Intraoperative blood loss (mL)	68.18 ± 6.37	43.16 ± 11.47	0.001

After one year of follow-up, successful fracture fixation was achieved in 27 patients (100%) in the lateral group and 24 patients (88.8%) in the posterolateral group. There was no loss of reduction in any patients in the lateral group, while three patients (11%) in the posterolateral group experienced loss of reduction. Although the loss of reduction was higher in the posterolateral group, it was not statistically significant (p=0.075). The fracture union time was similar between the two groups, with no statistically significant difference (12 weeks [range 10-16] vs. 10 weeks [range 8-14], p=0.3). Two patients in the posterolateral group developed deep surgical infections, while no infections were observed in the lateral group. However, this difference was not statistically significant (p=0.15) (Table 3).

Table 3 Comparison of functional outcomes between posterolateral and lateral approach groups

Outcome	Posterolateral group	Lateral group	*p*-value
Loss of reduction (%)	3 (11)	0 (0)	0.075
Fracture union time (weeks)	12 (range 10-16)	10 (range 8-14)	0.3
Deep surgical infection (%)	2 (7.4)	0 (0)	0.15

The functional outcomes, assessed using the IKDC, Lysholm, and Tegner scores, were significantly better in the lateral group. The IKDC score was significantly higher in the lateral group (92 ± 2.4 vs. 90 ± 1.3, p<0.001), as was the Lysholm score (96 ± 2.4 vs. 88 ± 1.5, p=0.001). The Tegner activity score was also significantly better in the lateral group (7 (6-8) vs. 6 (6-8), p=0.036) (Table 4).

Table 4 Comparison of IKDC, Lysholm, and Tegner scores between posterolateral and lateral approach groups

Outcome	Lateral group	Posterolateral group	*p*-value
IKDC score	92 ± 2.4	90 ± 1.3	0.001
Lysholm score	96 ± 2.4	88 ± 1.5	0.001
Tegner score	7 (6-8)	6 (6-8)	0.036

Neurological injury was observed in 2 patients (7.4%) in the posterolateral group, while no injuries were reported in the lateral group. Although neurological injury was more frequent in the posterolateral group, it was not statistically significant (p=0.15). Three cases of symptomatic implant issues were observed in the posterolateral group, while no such issues were reported in the lateral group (p=0.075).

The knee range of motion (ROM) after 12 months of follow-up was significantly better in the lateral group (130 degrees [range 110-135] vs. 125 degrees [range 90-130], p=0.001) (Table 5). Three patients in the posterolateral group exhibited an extension lag of 15-20 degrees, whereas no extension deficits were observed in the lateral group. All patients had full knee extension, and no clinical signs of knee instability were reported in either group.

Table 5 Comparison of knee range of motion between posterolateral and lateral approach groups

Knee Range of Motion	Lateral group	Posterolateral group	*p*-value
ROM (degrees)	130 (110-135)	125 (90-130)	0.002

Lastly, the degree of soft tissue damage, as indicated by less blood loss and shorter incision lengths, was significantly lower in the direct lateral approach group than the posterolateral approach.

Discussion

The clinical management of tibial plateau fractures, particularly posterolateral tibial plateau fractures (PLFs), requires careful surgical planning to optimize lower limb function and minimize postoperative complications. Various surgical techniques have been proposed to address the complexity of PLFs, but results vary significantly regarding functional recovery and postoperative outcomes. The present study underscores the effectiveness of the lateral approach for reducing and stabilizing depressed posterolateral tibial plateau fractures. This approach led to favorable clinical and radiological outcomes, suggesting that it could serve as an important technique in the armamentarium for tibial plateau fracture management. The primary objective in treating these fractures—achieving anatomical reduction without intra-articular step-offs—was successfully attained using the lateral approach, as supported by our findings and similar studies [8].

Previous research by Pean and Frosch [18,19] advocated for an extensive lateral approach with fibular osteotomy to gain exposure to the posterolateral tibial plateau. While effective, this method carries an increased risk of common peroneal nerve injury and disruption of the proximal tibiofibular joint. In contrast, the lateral approach used in the present study involves less invasive soft tissue dissection, leading to a shorter operative time and reduced intraoperative blood loss. These advantages align with the findings of Yu *et al.*, who demonstrated that limiting soft tissue damage can reduce the risk of neurovascular injury and improve recovery outcomes [20].

The importance of achieving precise articular reduction cannot be overstated. Zhai *et al.* highlighted that patients with greater than 2 mm of articular depression experience significantly higher rates of post-traumatic osteoarthritis compared to those with minimal depression [6]. The present study further supports these findings, as patients treated with the lateral approach demonstrated lower rates of postoperative articular collapse 12 months post-surgery, indicating that this approach may be more effective in preventing long-term joint degeneration. An additional benefit of the lateral approach is its minimal invasiveness, which was associated with lower rates of soft tissue complications and neurological injury. This is particularly important in light of previous studies, such as those by Solomon *et al.*, which reported increased rates of peroneal nerve injury with more invasive techniques like the posterolateral transfibular approach [17]. In the present study, the lateral approach did not result in any neurovascular injuries, which is a significant advantage over the posterolateral

methods. Functional outcomes, as measured by knee range of motion, IKDC scores, Lysholm scores, and Tegner activity scores, were significantly better in patients treated with the lateral approach. These findings are consistent with the work of He *et al.*, who demonstrated that modified approaches to PLFs can provide better joint visualization and outcomes, albeit with increased risks of neurovascular complications [18]. The present study demonstrates that the lateral approach offers a safe, efficient, and effective alternative to the more traditional posterolateral approach for the treatment of PLFs [19,20].

There are several limitations to consider. First, the relatively short follow-up period in this study may not be sufficient to evaluate long-term outcomes such as post-traumatic arthritis. Additionally, the sample size, while sufficient for initial evaluation, limits the generalizability of the findings to a broader population. Future studies with larger cohorts and extended follow-up periods will be critical in confirming the efficacy of the lateral approach for treating PLFs.

Conclusion

The results of this study showed that the modified lateral approach for depressed posterolateral tibial plateau fractures represents a significant advancement, achieving anatomical reduction and stable fixation with minimal soft tissue disruption, reduced operative time and blood loss, and a minimized risk of complications, including postoperative articular collapse and nerve injury, all while providing improved visualization through a limited incision. Despite these promising outcomes, the study's limitations, such as a small sample size and short follow-up, necessitate future research involving larger, multi-center cohorts with long-term follow-up to fully assess the risks of post-traumatic arthritis and hardware removal and to strengthen the evidence by placing greater emphasis on patient-reported outcomes.

Acknowledgements

The authors would like to thank all of patients who participated in this study.

Conflict of Interest

The authors declared no conflicts of interest in this work.

ORCID

Reza Noktehsanj
https://orcid.org/0000-0001-9008-4487
Ali Nami
https://orcid.org/0000-0002-1227-2818
Aziz Kamran
https://orcid.org/0000-0001-6065-4397

References

[1] Hu, S., Li, S., Chang, S., Du, S., Xiong, W. Surgical exposure to posterolateral quadrant tibial plateau fractures: an anatomic comparison of posterolateral and posteromedial approaches. *Journal of Orthopaedic Surgery and Research*, **2022**, 17(1), 346.

[2] Shen, Q.J., Zhang, J.L., Xing, G.S., Liu, Z.Y., Li, E.Q., Zhao, B.C., Zhang, T. Surgical treatment of lateral tibial plateau fractures involving the posterolateral column. *Orthopaedic Surgery*, **2019**, 11(6), 1029-1038.

[3] Liu, Z.Y., Zhang, J.L., Liu, C., Cao, Q., Shen, Q.J., Zhao, J.C. Surgical strategy for anterior tibial plateau fractures in hyperextension knee injuries. *Orthopaedic Surgery*, **2021**, 13(3), 966-978.

[4] Yoon, Y.C., Sim, J.A., Kim, D.H., Lee, B.K. Combined lateral femoral epicondylar osteotomy and a submeniscal approach for the treatment of a tibial plateau fracture involving the posterolateral quadrant. *Injury*, **2015**, 46(2), 422-426.

[5] Sohn, H.S., Yoon, Y.C., Cho, J.W., Cho, W.T., Oh, C.W., Oh, J.K. Incidence and fracture morphology of posterolateral fragments in lateral and bicondylar tibial plateau fractures. *Journal of Orthopaedic Trauma*, **2015**, 29(2), 91-97.

[6] Zhai, Q., Luo, C., Zhu, Y., Yao, L., Hu, C., Zeng, B., Zhang, C. Morphological characteristics of split-depression fractures of the lateral tibial plateau (Schatzker type II): A computer-tomography–based study. *International Orthopaedics*, **2013**, 37(5), 911-917.

[7] Xiang, G., Zhi-Jun, P., Qiang, Z., Hang, L. Morphological characteristics of posterolateral articular fragments in tibial plateau fractures. *Orthopedics*, **2013**, 36(10), e1256-e1261.
[8] Zhu, Y., Meili, S., Dong, M.J., Zhai, Q.L., Yao, L., Wang, J.C., Luo, C.F. Pathoanatomy and incidence of the posterolateral fractures in bicondylar tibial plateau fractures: A clinical computed tomography-based measurement and the associated biomechanical model simulation. *Archives of Orthopaedic and Trauma Surgery*, **2014**, 134(10), 1369-1380.
[9] Maroufi, P., Nazari, B. Density of One-Third Distal Radius in Patients Prone to Primary Osteoporosis, *Eurasian Journal of Science and Technology.* **2023**, 3(1), 23-28.
[10] Rudran, B., Little, C., Wiik, A., Logishetty, K. Tibial plateau fracture: Anatomy, diagnosis and management. *British Journal of Hospital Medicine*, **2020**, 81(10), 1-9.
[11] Giordano, V., do Amaral, N.P., Koch, H.A., e Albuquerque, R.P., de Souza, F.S., dos Santos Neto, J.F. Outcome evaluation of staged treatment for bicondylar tibial plateau fractures. *Injury*, **2017**, 48, S34-S40.
[12] Musahl, V., Tarkin, I., Kobbe, P., Tzioupis, C., Siska, P., Pape, H.C. New trends and techniques in open reduction and internal fixation of fractures of the tibial plateau. *The Journal of Bone & Joint Surgery British Volume*, **2009**, 91(4), 426-433.
[13] Hall, J.A., Beuerlein, M.J., McKee, M.D. Open reduction and internal fixation compared with circular fixator application for bicondylar tibial plateau fractures: surgical technique. *JBJS*, **2009**, 91(Supplement_2_Part_1), 74-88.
[14] Solomon, L.B., Stevenson, A.W., Baird, R.P., Pohl, A.P. Posterolateral transfibular approach to tibial plateau fractures: technique, results, and rationale. *Journal of Orthopaedic Trauma*, **2010**, 24(8), 505-514.
[15] He, X., Ye, P., Hu, Y., Huang, L., Zhang, F., Liu, G., Luo, C. A posterior inverted L-shaped approach for the treatment of posterior bicondylar tibial plateau fractures. *Archives of Orthopaedic and Trauma Surgery*, **2013**, 133(1), 23-28.
[16] Bernholt, D.L., DePhillipo, N.N., Grantham, W.J., Crawford, M.D., Aman, Z.S., Kennedy, M.I., LaPrade, R.F. Morphologic variants of posterolateral tibial plateau impaction fractures in the setting of primary anterior cruciate ligament tear. *The American Journal of Sports Medicine*, **2020**, 48(2), 318-325.
[17] Shuaishuai, W., Minglei, Z., Yue, Y., Dapeng, W., Tongtong, Z., Huimin, L. Clinical application of the modified posterolateral approach for treating posterior tibial plateau fractures. *Frontiers in Bioengineering and Biotechnology*, **2023**, 11, 1150541.
[18] Pean, C.A., Driesman, A., Christiano, A., Konda, S.R., Davidovitch, R., Egol, K.A. Functional and clinical outcomes of nonsurgically managed tibial plateau fractures. *JAAOS-Journal of the American Academy of Orthopaedic Surgeons*, **2017**, 25(5), 375-380.
[19] Frosch, K.H., Balcarek, P., Walde, T., Stürmer, K.M. A new posterolateral approach without fibula osteotomy for the treatment of tibial plateau fractures. *Journal of Orthopaedic Trauma*, **2010**, 24(8), 515-520.
[20] Noktehsanj, R., Movahedinia, M., Jousheghan, S.S., Salehi, B., Amini, K., Kamrava, S., Sajjadi, M.M. Clinical Outcomes of Fixation of Depressed Posterolateral Tibial Plateau Fractures Using a Direct Lateral Approach. *Indian Journal of Orthopaedics*, **2023**, 57(3), 466-471.

Copyright © 2026 by SPC (Sami Publishing Company) + is an open access article distributed under the Creative Commons Attribution License (CC BY), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.